

Matrix Theory

NS 2.4.1 Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by $T(a, b, c) = (b + c, c + a, a + b)$. Find $[T]_{C,B}$ in each of the following cases:

(a) $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$, $C = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$.

(b) $B = \{(1, 0, 0), (1, 1, 0), (1, 1, 1)\}$, $C = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$.

(c) $B = \{(1, 1, 2), (2, 1, 1), (1, 2, 1)\}$, $C = \{(2, 1, 1), (1, 2, 1), (1, 1, 2)\}$.

